

YAUVI Structural Workbench: evidence-bounded, reproducible structural protein analysis

Yuvraj Patel ¹

¹ Independent Researcher, Ohio, United States

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Researchers often need to assess several kinds of evidence before interpreting a protein structure. YAUVI Structural Workbench brings those records into a local Python suite and browser interface for six structural-protein questions: coordinate identity and quality, membrane orientation, conformational resemblance, functional-site context, biological assembly interfaces, and comparative structural/sequence relationships. Each analysis remains an independently installable command-line package. The workbench creates typed configurations, verifies files by SHA-256, invokes the registered command, and renders its JSON and tabular evidence into a readable, printable report.

The central design constraint is that evidence dimensions are not collapsed into a universal protein or function score. Unknown provenance remains unknown; active-like describes coordinate resemblance rather than biochemical activity; an annotated catalytic site is not observed catalysis; and a shared fold is not exact functional transfer. Missing references, mappings, runtimes, or validation reports remain visible as missing or scientifically incomplete evidence.

Statement of need

Structural bioinformatics typically requires moving between coordinate validation, residue numbering systems, sequence annotations, reference conformations, biological assemblies, surface areas, and similarity searches. Author and label residue identities in PDBx/mmCIF are particularly important when a measurement must trace back to an exact sequence position. Community validation systems such as wwPDB and MolProbity provide essential model-quality evidence ([Williams et al., 2018](#); [Worldwide Protein Data Bank, n.d.](#)), while Gemmi provides robust macromolecular format support ([Wojdyr, 2022](#)). However, these results are often reviewed in separate tools, and missing evidence can be mistaken for a favorable or negative result.

YAUVI provides one reproducible case boundary while leaving scientific calculations in standalone packages. It is intended for structural biologists, bioinformaticians, and trainees who need inspectable evidence and explicit claim limits rather than an opaque ranking. The same deterministic reports can support method development, target characterization, comparison of experimental and predicted structures, and planning of follow-up validation without generating a wet-lab protocol.

State of the field and build-versus-contribute justification

Existing programs already provide stronger specialist calculations than a new implementation should attempt to replace. Gemmi supplies standards-aware coordinate parsing ([Wojdyr, 2022](#));

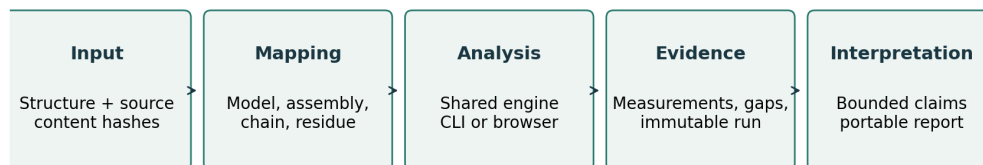
39 wwPDB and MolProbity supply community model validation (Williams et al., 2018; Worldwide
40 Protein Data Bank, n.d.); OPM/PPM supplies a richer membrane-positioning reference system
41 (Lomize et al., 2006); and Foldseek and DIAMOND supply established structure and sequence
42 searches (Buchfink et al., 2021; Kempen et al., 2024). YAUVI therefore invokes or imports
43 these methods through named adapters and preserves their scores, parameters, units, versions,
44 and limitations rather than reimplementing them.

45 Broader libraries already integrate important parts of this workflow. Biotite combines
46 sequence and structure representations, database access, and interfaces to external applications
47 (Kunzmann & Hamacher, 2018). ProDy provides established tools for protein structural
48 dynamics and ensemble analysis (Bakan et al., 2011). These are relevant foundations and
49 alternatives for users building their own analyses. YAUVI's present comparison is architectural;
50 no benchmark establishing superior speed or accuracy over these packages has been performed.

51 The contribution proposed here is an explicit case and reporting contract: exact input identity,
52 residue mapping, explicit provenance, fail-closed preflight, separation of scientific dimensions,
53 deterministic reports, and claim ceilings shared across six independently runnable workflows.
54 A separate workbench is intended to make these contracts usable across several engines and
55 accessible to researchers who do not write Python. This choice adds packaging and adapter
56 maintenance; reusable numerical or parsing improvements should still be contributed upstream
57 where appropriate.

58 Software design

A shared path from input to interpretation



Execution outcome • Scientific qualification • Independent reproduction • Publication approval

Four separate states; success in one does not establish the others.

Figure 1: Shared execution and provenance architecture. Four status dimensions remain separate.

59 The workbench stores input bytes content-addressably and creates immutable run records.
60 Browser actions and command-line commands call the same case service, which keeps the
61 interface from becoming a second scientific implementation. Parameter edits preserve earlier
62 case revisions. A bounded local worker records queued and running jobs; server restarts expose
63 interrupted work for deliberate resubmission. Failed attempts remain inspectable and cannot
64 satisfy a completed-run cache lookup. Reuse requires matching input, source-code, parameter,
65 and runtime identities, together with intact output checksums. Task definitions declare the
66 scientific question, accepted artifact types, format validator, source assistance, missing-evidence
67 consequence, outputs, and claim ceiling. A source finder links input roles to official RCSB
68 PDB, wwPDB validation, AlphaFold DB, UniProt, SIFTS, M-CSA, PDB CCD, ChEBI, and
69 OPM/PPM resources. Network access is disabled by default. When explicitly enabled, only
70 registered artifact types and public identifiers can be acquired; cache acquisition and adoption
71 into an analysis are separate operations.

72 StructQC establishes coordinate provenance and residue identity before composed workflows.
73 Modified chemical components are preserved separately from explicit parent-residue sequence

normalization; normalization does not imply identical chemistry. SIFTS provides a reference resource for sequence-to-structure mapping (Velankar et al., 2013). MembraneOrient separates experimental beta-barrel and alpha-helical helix-axis paths, with OPM/PPM retained as an external comparison standard (Lomize et al., 2006). StateAtlas uses exact declared residue equivalences for the candidate ABL-family Mark 1 scope, Kabsch alignment, RMSD/RMSF, deterministic clustering, and two-sided experimental references. SiteContext and ActState keep annotation, observed chemistry, and geometric competence separate. Declared cofactors and observed non-solvent groups cannot establish occupancy without exact component identity and site proximity; that evidence remains unavailable where the current ActState reader cannot establish it. M-CSA annotations support curated site interpretation and retain their evidence boundaries (Ribeiro et al., 2018). AssemblyContext reports heavy-atom contacts, stoichiometry evidence, and method-specific solvent accessible surface area, including named FreeSASA calculations (Mitternacht, 2016). SF-CSA executes Foldseek (Kempen et al., 2024) and DIAMOND (Buchfink et al., 2021) as separate structural and sequence legs against checksum-pinned reference universes.

Completed or scientifically incomplete runs emit `REPORT_DATA.json`, printable `REPORT.html`, a deterministic `RAW_EVIDENCE.zip`, checksums, and the canonical run manifest. Display rounding never modifies the underlying scientific values.

Validation and research use

The suite uses synthetic offline fixtures, schema tests, fail-closed boundary tests, transformation and ordering invariance tests, deterministic output comparisons, controller security tests, and standalone package tests. External scientific qualification is reported separately from software correctness. The first checksum-locked public collection includes wwPDB validation, OPM membrane strata, KinCore-labeled two-sided conformational references, an M-CSA enzyme case, a deposited assembly evaluated with FreeSASA, and a CATH-labeled SF-CSA mini-database searched by Foldseek and DIAMOND. Four public cases pass their predeclared gates and two remain partial. Qualification v2 separately freezes scope-specific strata, development and held-out splits, evidence requirements, and unchanged gates. The reviewed collection 2.9 baseline contains five executed panels. Coordinate quality, functional-site context, and assembly interfaces each passed 16 cases; ABL state comparison passed four reference cases and ten held-out cases, with reference classifications informed by KinCore (Modi & Dunbrack, 2022). These four blocking panels produced identical case verdicts across six recorded operating-system and Python combinations, including four additional control outcomes per runner. Membrane orientation passed 5/16 cases in five runners and 4/16 in one; it remains experimental. SF-CSA's sixteen required cases were unexecuted following withdrawal of changing live-query proteome inputs. These are historical measurements, not an accuracy estimate for the toolkit or qualification of subsequent code changes.

Recent hardening exposed a reproduction-checker defect: matching failing panels could produce aggregate success. The replacement checks required workflow coverage, input and protocol identities, exact case IDs, controls, and per-case verdicts. A missing panel, incompatible record, or insufficient environment count prevents release qualification. The distinction between software tests, cross-environment execution, independent scientific review, and human release authorization remains explicit. No scope is claimed independently qualified for the changed build.

Historical collection 2.9 · changed code requires fresh qualification

Workflow	Cases in compared panel	Historical result	Interpretation
StructQC	16	16/16	Curated case execution
SiteContext	16	16/16	Curated case execution
AssemblyContext	16	16/16	Curated case execution
ABL StateAtlas	14: 4 refs + 10 held-out	14/14	Curated case execution
SF-CSA	16	Unexecuted	Pending source recovery
MembraneOrient	16	5/16	Experimental

Six archived runners: four required panels agree; membrane is 5/16 in five and 4/16 in one. Controls are separate (four per runner). Unexecuted alpha-helical requirements are not shown as tested cases. Small curated panels do not establish broad biological validity; no combined accuracy score.

Figure 2: Historical qualification evidence. Counts describe curated panels and retain reference, experimental and unexecuted distinctions; they are not toolkit accuracy.

118 Research impact statement

119 During public development, the project has not recorded independent adoption, published
 120 research use, or five completed external benchmark gates. The software is therefore not
 121 presented as JOSS submission-eligible. Current reproducible benchmark records expose both
 122 successful cases and scientific limitations; they do not substitute for documented use in
 123 real structural-biology analyses, independent installation feedback, and public issue-driven
 124 refinement. Aspirational utility is not counted as realized research impact in the release state.

125 Limitations

126 The workbench does not replace experimental validation, density inspection, biochemical
 127 assays, native-surface measurements, or expert curation of conformational references. Optional
 128 external executables and databases retain their own licenses and must be acquired independently.
 129 Predicted models cannot establish activity, assemblies may be context-dependent, solvent-
 130 accessible area is method-specific, and sequence or structure similarity alone cannot transfer
 131 mechanism.

132 AI usage disclosure

133 OpenAI Codex assisted with implementation, tests, interface text, documentation, and
 134 manuscript drafting. Anthropic Claude assisted during earlier development. Exact
 135 model/version reconciliation remains incomplete and must be resolved before submission.
 136 Earlier drafts record human review; review of the current changes remains pending. Automated
 137 tests do not substitute for that review. The human author remains responsible for originality,
 138 accuracy, licensing, ethical and legal compliance, and all claims. AI output is not scientific
 139 evidence. The author will handle editor/reviewer conversations without AI assistance, except
 140 translation where journal policy permits it.

141 Conflicts of interest and funding

142 Conflict-of-interest and funding statements have not yet been finalized for submission. They
 143 must be supplied and approved by every listed author before the paper can be submitted.

Acknowledgements

The project depends on the maintainers and data curators of its open scientific dependencies and public reference resources. Those projects must be cited independently when their methods or data are used.

References

- Bakan, A., Meireles, L. M., & Bahar, I. (2011). ProDy: Protein dynamics inferred from theory and experiments. *Bioinformatics*, 27(11), 1575–1577. <https://doi.org/10.1093/bioinformatics/btr168>
- Buchfink, B., Reuter, K., & Drost, H.-G. (2021). Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nature Methods*, 18, 366–368. <https://doi.org/10.1038/s41592-021-01101-x>
- Kempen, M. van, Kim, S. S., Tumescheit, C., Mirdita, M., Lee, J., Gilchrist, C. L. M., Söding, J., & Steinegger, M. (2024). Fast and accurate protein structure search with foldseek. *Nature Biotechnology*, 42, 243–246. <https://doi.org/10.1038/s41587-023-01773-0>
- Kunzmann, P., & Hamacher, K. (2018). Biotite: A unifying open source computational biology framework in python. *BMC Bioinformatics*, 19, 346. <https://doi.org/10.1186/s12859-018-2367-z>
- Lomize, A. L., Pogozheva, I. D., Lomize, M. A., & Mosberg, H. I. (2006). Positioning of proteins in membranes: A computational approach. *Protein Science*, 15(6), 1318–1333. <https://doi.org/10.1110/ps.062126106>
- Mitternacht, S. (2016). FreeSASA: An open source c library for solvent accessible surface area calculations. *F1000Research*, 5, 189. <https://doi.org/10.12688/f1000research.7931.1>
- Modi, V., & Dunbrack, R. L., Jr. (2022). Kincore: A web resource for structural classification of protein kinases and their inhibitors. *Nucleic Acids Research*, 50, D654–D664. <https://doi.org/10.1093/nar/gkab920>
- Ribeiro, A. J. M., Holliday, G. L., Furnham, N., Tyzack, J. D., Ferris, K., & Thornton, J. M. (2018). Mechanism and catalytic site atlas (m-CSA): A database of enzyme reaction mechanisms and active sites. *Nucleic Acids Research*, 46, D618–D623. <https://doi.org/10.1093/nar/gkx1012>
- Velankar, S., Dana, J. M., Jacobsen, J., Ginkel, G. van, Gane, P. J., Luo, J., Oldfield, T. J., O'Donovan, C., Martin, M.-J., & Kleywegt, G. J. (2013). SIFTS: Structure integration with function, taxonomy and sequences resource. *Nucleic Acids Research*, 41, D483–D489. <https://doi.org/10.1093/nar/gks1258>
- Williams, C. J., Headd, J. J., Moriarty, N. W., Prisant, M. G., Videau, L. L., Deis, L. N., Verma, V., Keedy, D. A., Hintze, B. J., Chen, V. B., Jain, S., Lewis, S. M., Arendall, W. B., Snoeyink, J., Adams, P. D., Lovell, S. C., Richardson, J. S., & Richardson, D. C. (2018). MolProbity: More and better reference data for improved all-atom structure validation. *Protein Science*, 27(1), 293–315. <https://doi.org/10.1002/pro.3330>
- Wojdyr, M. (2022). Gemmi: A library for structural biology. *Journal of Open Source Software*, 7(73), 4200. <https://doi.org/10.21105/joss.04200>
- Worldwide Protein Data Bank. (n.d.). *wwPDB validation reports*. <https://www.wwpdb.org/validation/validation-reports>